



AC Magnetic Loss Reduction of Fe-(x)Si Soft Magnetic Composites

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Abstract

The aim of this study was to improve the magnetic properties and reduce the core loss of a soft magnetic composite (SMC) core as the mixture of Fe–Si alloy and epoxy. In order to develop a soft magnetic core with high power driving characteristics, we investigated the magnetic and loss characteristics at different Si contents in the Fe–Si alloy-based SMC core. As the Si content was increased from 3.5 to 6.5 wt.%, the permeability and Q-factor of the core increased, and the core loss decreased. In particular, the Fe-6.5Si alloy-based SMC core containing 6.5 wt.% of Si showed the highest permeability of 23.27 H/m at 1 MHz as well as the largest Q-factor of 66.58 at the resonance frequency of 1.38 MHz. The Fe-6.5Si SMC core exhibited the lowest core loss of 693.94 mW/cm³ at 1 MHz. The findings of this study showed that the changes in the magnetic and electrical properties of the powder due to the addition of Si affects the magnetic permeability and core loss properties of the SMC core.

Keywords Soft magnetic composites · Magnetic permeability · Core loss

1 Introduction

The recent boom in the use of mobile devices, such as smartphones and laptops, has highlighted the importance of miniaturized and highly efficient electronic components for manufacturing portable products with long driving times. Moreover, the application field of Internet of Things is gradually expanding, and the commercialization of electric vehicles is slowly becoming viable because of the growing interest in eco-friendly technologies. As a result, high output of devices is also required as a core technology, and thus, research on the development of miniaturized, highly efficient, and high-output electronic components is still continuing worldwide [1–3]. Among the various candidate materials being considered for developing high-efficiency and high-output devices, soft magnetic materials are widely used in many electronic components that exhibit a high saturation magnetization (M_s), low coercivity (H_c), and low core loss. In particular, soft magnetic composites (SMCs) manufactured by mixing soft magnetic materials with polymer binders are widely used in power components, such

as inductors and transformers [4–9]. Studies are underway on Fe–Si alloys [10–12], Fe–Ni alloys [13, 14], Fe–Si–Cr [15–18], Fe–Si–Al [19–21], and amorphous Fe [22–24] based SMCs.

The core technology required for the aforementioned next-generation electronic devices is centered on the properties of soft magnetic materials. First, as evident from Eq. (1), the inductance of soft magnetic materials may be maintained by improving the permeability of the material for miniaturization of the component.

$$L = \mu_r \mu_0 \frac{AN^2}{l} \quad (1)$$

Here, L is the inductance, N is the number of windings of the coil, A is the cross-sectional area of the core, l is the length of the coil, μ_r is the relative permeability, and μ_0 is the permeability in vacuum. Second, the core loss is an important factor for evaluating the energy efficiency of electronic components, including inductors. In particular, as the driving frequency of electronic components is gradually increased, the low core loss in the high frequency band of 1 MHz or more becomes noticeable. Core loss (P_c) is the sum of hysteresis loss (P_h), eddy-current loss (P_e), and residual loss (P_r), as shown in Eq. (2) below [25, 26].

$$P_c = P_h + P_e + P_r \cong P_h + P_e \quad (2)$$

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P_h is proportional to BH_{max} , which is the area of the magnetic hysteresis loop, and to reduce this loss, soft magnetic materials with low H_c are mainly used. P_e is composed of an intra-particle eddy-current loss that occurs inside the material and an inter-particle eddy-current loss generated by the contact between the materials. The intra-particle eddy-current loss can be reduced by increasing the resistivity of the material, whereas the inter-particle eddy-current loss can be reduced by coating of the material with an insulator. P_r is negligible, and hence, can be ignored in the case of metal composites. In our case, P_c was considered to be composed of only P_h and P_e [25].

In this study, we used an Fe–Si-based alloy as the soft magnetic material. The magnetic properties of the core and the changes in the core loss were examined by changing the ratio of Si from 3.5 to 6.5 wt.%. When Si was alloyed with Fe, the electrical resistivity of the material increased more than that of the material containing pure Fe. This increased electrical resistivity can reduce the eddy-current loss and increasing the thermal stability of the material.

2 Experimental Details

Crystalline Fe–Si alloy powders with different Si contents of 3.5, 4.5, and 6.5 wt.% (hereafter, Fe-3.5Si, Fe-4.5Si, and Fe-6.5Si) were manufactured by gas atomization. To minimize the changes in the electrical and magnetic characteristics of the SMC core induced by the change in the size of the Fe-(x)Si powder, the prepared powders were sieved using a standard net (#400, mesh size: $\sim 38 \mu\text{m}$).

A toroidal core was manufactured to measure the magnetic properties and core loss of the prepared crystalline metal-based SMCs. For the toroidal core, 2.0 wt.% epoxy resin was added to 10 g of the powder sample and mixed in an agate mortar for 30 min to 1 h. The prepared mixture was then transferred to a cylindrical mold and pressed to form a cylindrical toroidal core with inner and outer diameters of 13 and 20 mm, respectively. The compression pressure was fixed at 1.0 ton/cm^2 . The prepared core was heat-treated in air at 160°C for 1 h, and then naturally cooled to room temperature.

X-ray diffraction (XRD; Rigaku Miniflex) using $\text{Cu-K}\alpha_1$ radiation (wavelength, $\lambda = 1.5405 \text{ \AA}$) was used to analyze the crystal phase of the prepared powders. Scanning electron microscopy (SEM; Hitachi S-4800) and energy-dispersive spectroscopy (EDS) were employed to analyze the morphology and composition of the prepared powder. A particle size analyzer (PSA, Microtrac S3500) was used to deduce the particle size distribution and average particle size of the powder. The electrical resistivity of the powder was measured using a powder resistivity measurement system. The DC magnetic properties of the powder and prepared core were

determined by a vibrating sample magnetometer (Lakeshore VSM7404) in the range of -1 T to $+1 \text{ T}$. The core permeability and Q-factor were measured using an impedance analyzer (Agilent 4294A), a 4294A adapter, and a 16902A test fixture in the 100 kHz to 110 MHz band. For these measurements, an insulator-coated Cu wire with a diameter of 4.5 mm was wound 10 times along the circumference of the core. To measure the core loss, both the primary and secondary coils were wound 30 times along the circumference of the core using an Ni wire with a diameter of 0.25 mm. The core loss was measured by a B-H analyzer (Iwatsu SY-8218). The maximum magnetic flux density (B_m) was fixed at 20 mT, and the changes in the core loss were measured in the 100 kHz to 1 MHz frequency band.

3 Results and Discussion

Figure 1 shows the XRD patterns of the Fe-(x)Si powders prepared by gas atomization. Based on the ICDD database ICDD #06–0696, it was confirmed that all the diffraction patterns of the measured Fe-(x)Si alloy exactly match those of the α -Fe phase. In addition, as the content of Si was from 3.5 to 6.5 wt.%, the position of the main peak gradually increased from 44.76° to 44.78° and 44.87° , and all the observed peaks shifted to high 2θ values. This implies that Si ($1.18 \text{ \AA}$) atoms with atomic radii smaller than that of Fe

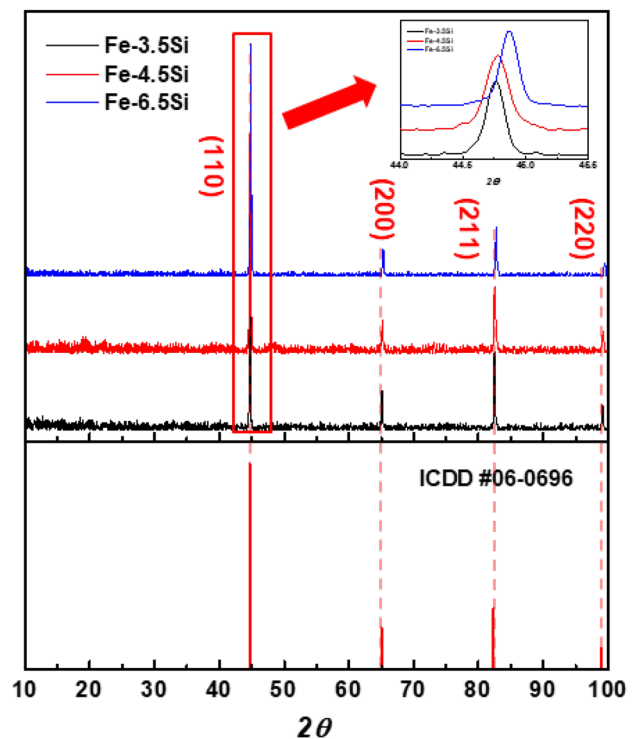


Fig. 1 XRD patterns of Fe-(x)Si

(1.24 Å) were accurately replaced in the α -Fe cubic structure matrix, thereby reducing the lattice constant [27].

The SEM image of the prepared powder and the results of EDS component analysis were shown in Fig. 2. The SEM image confirms that the prepared powder particles were spherical and have uniform sizes. The EDS composition analysis showed that the Fe-(x)Si alloy powder did not contain any other element except Fe and Si, and thus, confirmed its high purity. The content of Si was also confirmed with an error of 1 wt.%.

Figure 3 shows the particle size analysis results of the sieved powders. All the Fe-(x)Si alloys showed particle size distributions in the range of 62–70 μm . The D50 values representing the average particle size of the measured powder were found to be 26.75, 20.54, and 18.10 μm for Fe-3.5Si, Fe-4.5Si, and Fe-6.5Si, respectively. In particular, the difference in the average particle size of the Fe-(x)Si alloys was confirmed to be 10 μm or less, which may minimize the effect of the changes in the electrical and magnetic characteristics of the powders on the magnetism and loss characteristics of the applied SMC core.

The resistivity of the powder is an important physical parameter that directly affects the intra-particle eddy-current loss of the SMC core. Figure 4 shows the resistivity of the powder measured using the powder resistance measurement system. The measured resistivity of the powder increased rapidly as the Si content was increased at all the applied pressures. The change in the resistivity according to the applied pressure showed a decreasing trend as the pressure was increased. In particular, the estimated resistivities at 1.8

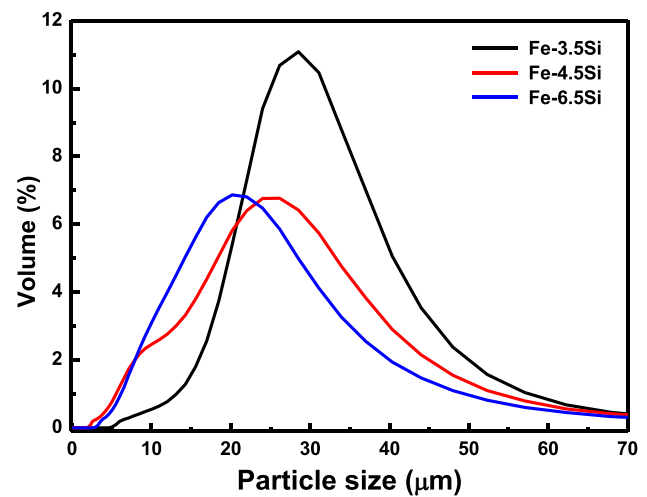


Fig. 3 Particle size distribution of Fe-(x)Si

ton ($\sim 1.0 \text{ ton/cm}^2$), which was the basic molding pressure of the core, were approximately 200, 700, and 3.23 $\text{m}\Omega\bullet\text{cm}$ for Fe-3.5Si, Fe-4.5Si, and Fe-6.5Si, respectively. Therefore, among the prepared Fe-(x)Si powders, Fe-6.5Si is expected to exhibit the highest electrical resistivity, and thus, the lowest eddy-current.

Figure 5a shows the magnetic hysteresis loop of Fe-(x) Si powder measured at room temperature. As evident, all the Fe-(x)Si showed good soft magnetic properties with high M_s and low H_c . Figure 5b reveals the changes in M_s as the Si content in the powder is increased, and clearly,

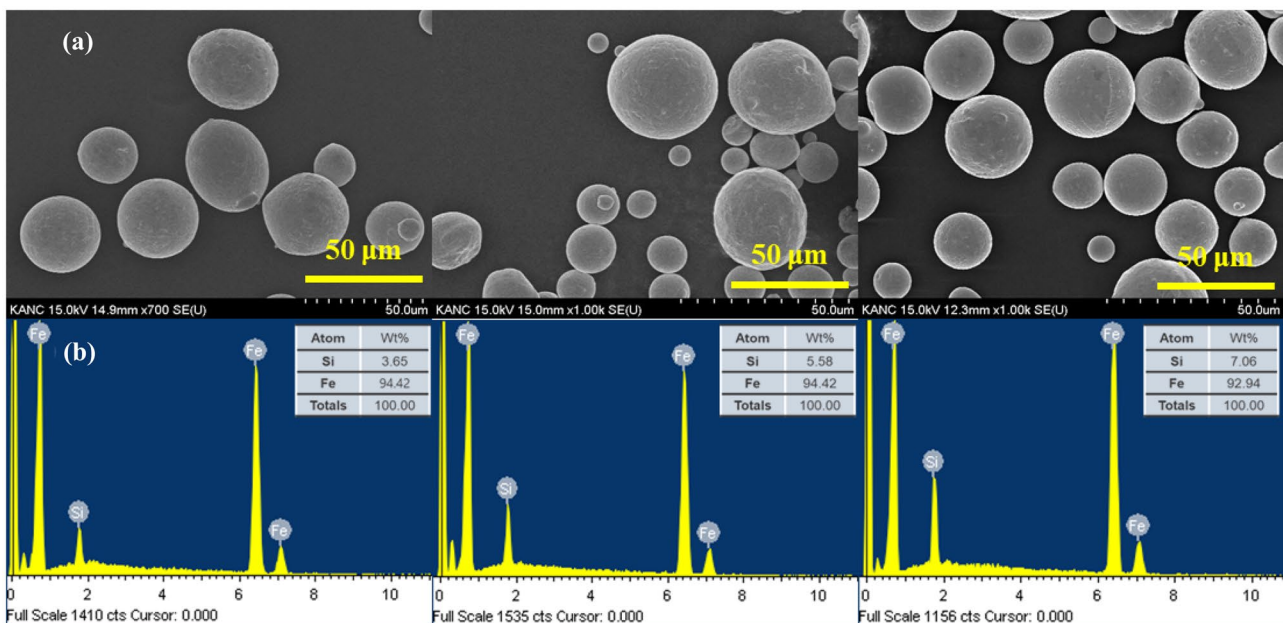


Fig. 2 a Morphology and b EDS analyses results of Fe-(x)Si

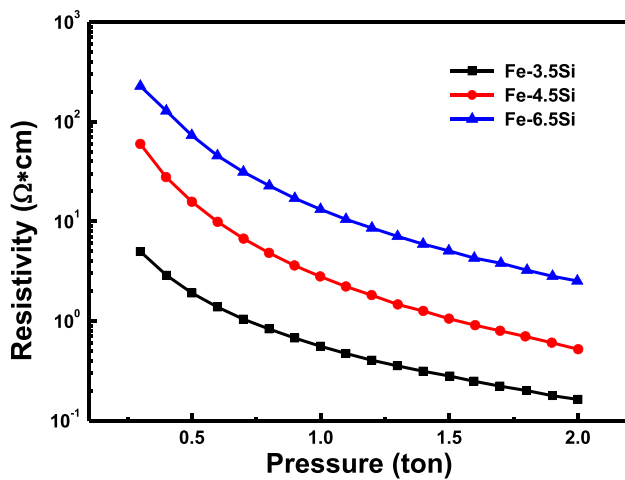


Fig. 4 Variations in powder resistivity according to the changes in the pressure applied on the Fe-(x)Si powders

M_s shows a tendency to decrease with the increasing Si concentration in the powder samples. The decrease in M_s is caused by the decreasing concentration of Fe, which is magnetic, as a result of the increasing concentration of Si, which is non-magnetic, in the powders.

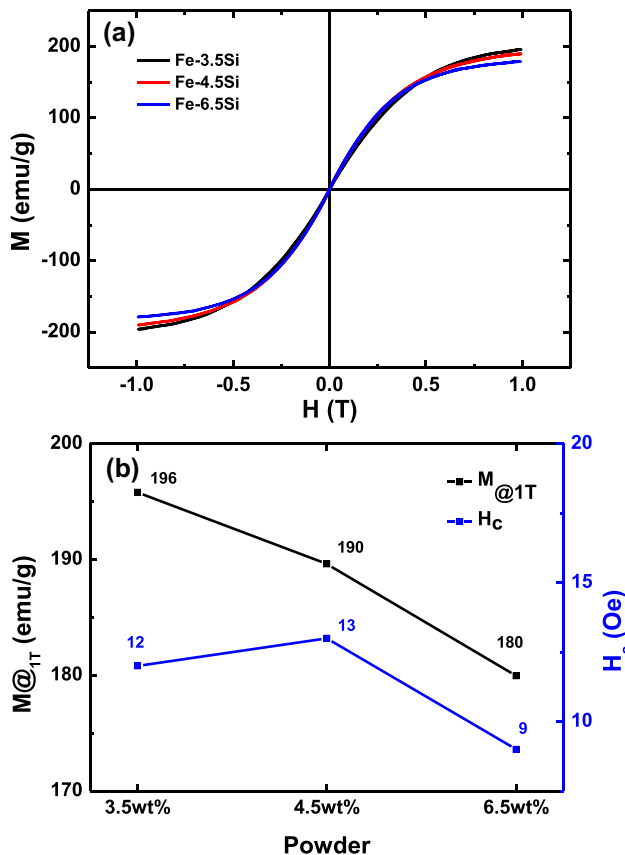


Fig. 5 **a** Magnetic hysteresis loop, and **b** variations in M_s and H_c of Fe-(x)Si

Figure 6 shows the variations in the permeability and Q-factor of the SMC core with the increasing Si content. As shown in Fig. 6a, the magnetic resonance frequency of the core increases as the Si content increases. The permeability of the core was between 23 and 24 H/m in all the bands below the magnetic resonance frequency. Additionally, no significant changes were observed in the permeability with Si addition in the frequency band below 1 MHz. However, above this frequency band, noticeable changes in the permeability were observed with the increasing Si content in the powders. The Q-factor representing the performance of the core also showed a clear correlation with the Si content, as shown in Fig. 6b. As the Si content was increased from 3.5 to 6.5 wt.%, the Q_{max} value increased from 42 to 66 in all the measured frequency bands, and the frequency band representing the Q_{max} value also increased from 1 MHz or less to around 2 MHz.

Figure 7 shows the loss analysis results of the SMC core. Figure 7a indicates that the core loss, represented by P_c , decreased in all the measured frequency bands as the Si content was increased. At a frequency of 1 MHz, the P_c of the prepared powders, i.e., Fe-3.5Si, Fe-4.5Si, and

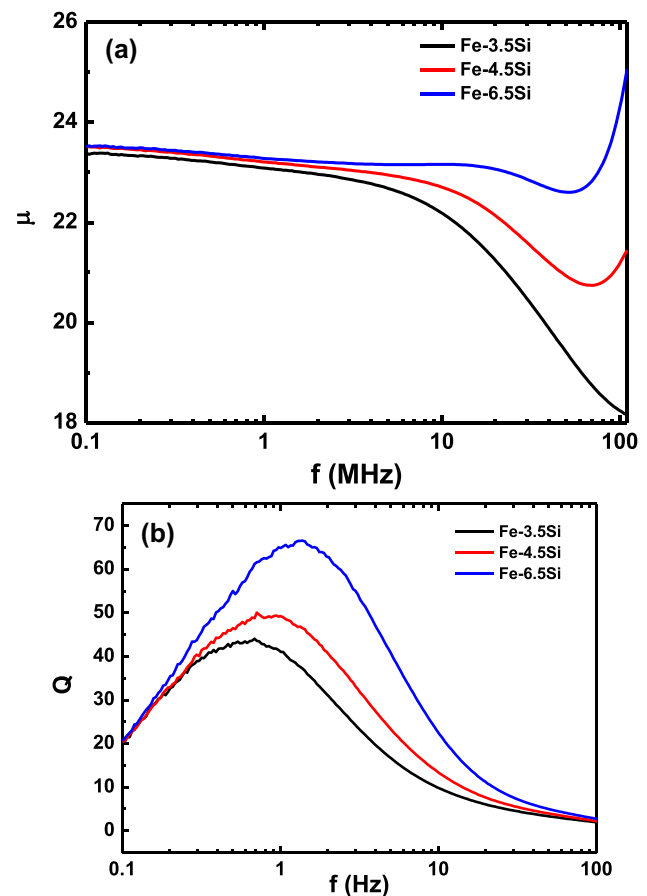


Fig. 6 Changes in the **a** permeability, and **b** Q-factor according to the Si content in the Fe–Si alloy-based SMC core

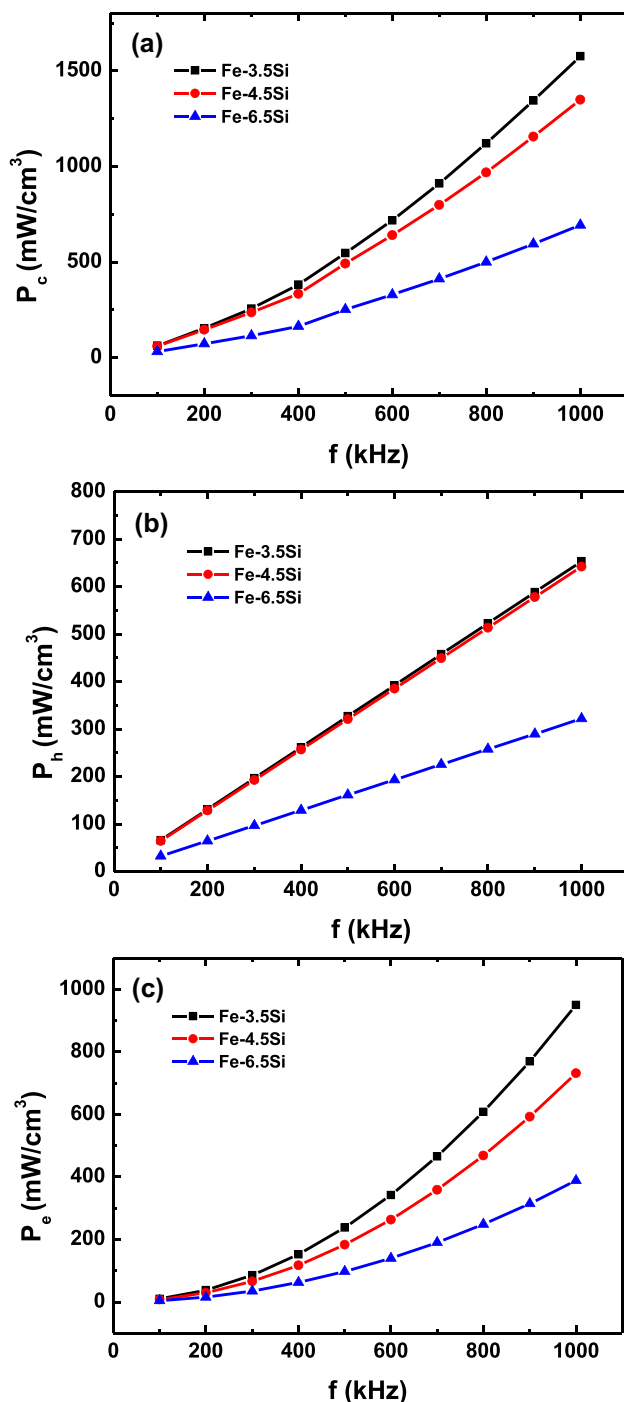


Fig. 7 **a** Total core loss, **b** hysteresis loss, and **c** eddy-current loss of Fe-(x)Si SMC materials as a function of measured frequency

Fe-6.5Si, were measured as 1576, 1349.76, and 694 mW/cm³, respectively, indicating that the sample Fe-3.5Si exhibited the highest core loss at 1 MHz, and that the total core loss decreased significantly with the increasing Si content. As shown in Fig. 7b, c, the Fe-3.5Si- and Fe-4.5Si-based SMC cores showed very similar P_h values, whereas the P_e

values of these cores decreased with the increasing Si content, similar to their P_c values. These results imply that in Fe–Si–based SMC cores, any change in the P_e due to an increase in the Si content in the alloy powder produces a remarkable change in the P_c of these cores; thus, the total core losses in Fe–Si–based SMC cores are majorly governed by the intra- and inter-particle eddy-current losses. In addition, it was confirmed that the Fe-6.5Si alloy-based SMC core exhibited the highest permeability, the largest Q-factor, and the lowest core loss.

4 Conclusion

In this study, the permeability and core loss characteristics of Fe–Si alloy-based SMC cores were evaluated by varying the Si content in the alloy powder. The analysis results confirmed that as the Si content was increased from 3.5 to 6.5 wt.%, the magnetic permeability of the core increased, whereas the core loss decreased. Among the prepared samples, the Fe-6.5Si alloy-based SMC core with a Si content of 6.5 wt.% showed the highest permeability of 23.27 H/m, the largest Q-factor of 66.58, and the lowest core loss of 693.94 mW/cm³ at 1 MHz. The model presented here to elucidate the changes in magnetism and core loss of Fe–Si-based SMC cores according to the Si content of the alloy is anticipated to be useful for controlling and implementing specific properties required in electronic components or parts in the future.

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Declarations

Conflict of Interest The authors declare no competing interests.

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